NCTech applies four layers of IT security

1. **Human layer**

   People, both internal and external to NCTech, present a significant risk to the integrity of our IaaS business model. NCTech staff should fully understand the complexity of the technology in use and how it can remain secure.

   Security controls applied at this layer

   – Robust access control procedures help to reduce the risk of a security breach.

   – Compulsory training ensures compliance with NCTech's standard operating procedures.

2. **Perimeter layer**

   The perimeter is the outer layer of the whole NCTech system and is the point where clients connect and gain access to their own data and applications. This layer includes all client devices which are connected to the NCTech system, so this could potentially be the most vulnerable part of our system.

   Security controls applied at this layer

   – All external client devices need to be registered and authorised to gain access to the IaaS system. This is to ensure that only recognised and trusted devices can cross the system perimeter.

   – A unique security key is allocated for each client device.

3. **Application layer**

   The application layer relates to the software we use to provide services to our clients. This includes the IaaS operating system as well as the varied applications used by our clients. The application layer enables the whole system to function effectively and securely, within prescribed operating parameters.

   Security controls applied at this layer

   – All clients' applications need to be checked, formally approved, and registered to prevent the use of unauthorised and unsafe software applications on the IaaS system.

4. **Data layer**

   Client data held and processed on our system is the most likely target for cyber-attacks, so the integrity of client data must be a fundamental purpose of our security measures. Any data loss will erode client trust in our business model and could result in a significant downturn in business activity, with a consequent deterioration in financial performance. Data breaches could also expose the company to regulatory fines and other legal sanctions.

   Security controls applied at this layer

   – Firewalls to control the flow of incoming and outgoing data on the IaaS system.

   – Vigorous procedures to ensure the secure access and integrity of client data held on the IaaS system.